SIMATS ENGINEERING

SIMATS Online Programming Lab

C++ PROGRAMMING

JANA LAHARI
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Assignment_01_CO_02

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QUESTIONS

- Q1 Check whether a number is a sunny number — 5 marks
- Q2 Check whether a number is a fascinating number — 5 marks
- Q3 Check whether a number is a Disarium number — 5 marks
- Q4 Check whether a number is a Kaprekar number — 5 marks
- Q5 Check whether a number is a Keith number — 5 marks
- Q6 Check whether a number is a Trimorphic number — 5 marks

6/6 answered

Q1 Check whether a number is a sunny number — 5 marks

Topics: Classes, objects, member functions, loops and mathematical logic.

Question: A sunny number is a number whose successor is a perfect square. Create a SunnyNumber class to determine whether a number satisfies this condition.

Example: 8 + 1 = 9 = 3^2
Test cases:
1.8->Sunny number
2.3->Sunny number
3.10->Not a sunny number

Your Code

```cpp
#include <iostream>
#include <cmath>
using namespace std;
class SunnyNumber {
private:
    int num;
public:
    SunnyNumber(int n) {
        num = n;
    }
    bool isSunny() {
        int nextNum = num + 1;
        int sqRoot = round(sqrt(nextNum));
        return (sqRoot * sqRoot == nextNum);
    }
};

int main() {
    int n;
    if (cin >> n) {
        SunnyNumber obj(n);
        if (obj.isSunny()) {
            cout << "Sunny number" << endl;
        } else {
            cout << "Not a sunny number" << endl;
        }
    }
    return 0;
}
```

Input

```
8
```

Output

```
Sunny number
```

Visible Test Cases

- Test Case 1 ✓
- Test Case 2 ✓
- Test Case 3 ✓

5.00 / 5 marks

Submitted

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6/6 answered

Q2 Check whether a number is a fascinating number — 5 marks

Topics: Classes, member functions, loops, digit extraction, logical validation.

A three-digit number is fascinating if multiplying it by 1, 2, and 3 and concatenating the results produces each digit from 1 to 9 exactly once. Create a NumberAnalyzer class to check the condition.

Example: 192 → 192, 384, 576; together they contain digits 1–9 exactly once.
Test cases:
1.192->Fascinating number
2.219->Fascinating number
3.123->Not a fascinating number

Your Code

```cpp
#include <iostream>
#include <string>
#include <algorithm>
using namespace std;
class NumberAnalyzer {
private:
    int num;
public:
    NumberAnalyzer(int n) {
        num = n;
    }
    bool isFascinating() {
        string concatenated = to_string(num) + to_string(num * 2) + to_
        if (concatenated.length() != 9) {
            return false;
        }
        sort(concatenated.begin(), concatenated.end());
        string expected = "123456789";
        return (concatenated == expected);
    }
};
int main() {
    int n;
    if (cin >> n) {
        NumberAnalyzer analyzer(n);
        if (analyzer.isFascinating()) {
            cout << "Fascinating number" << endl;
        } else {
            cout << "Not a fascinating number" << endl;
        }
    }
    return 0;
}
```

Input

```
192
```

Output

```
Fascinating number
```

Visible Test Cases

- Test Case 1 ✓
- Test Case 2 ✓
- Test Case 3 ✓

5.00 / 5 marks

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6/6 answered

Q3 Check whether a number is a Disarium number — 5 marks

Topics to use: Classes, objects, private data members, public member functions, loops, digit extraction, mathematical calculations.

Question: A number is called a Disarium number if the sum of its digits raised to their respective positions equals the original number. The position count starts from the leftmost digit.

Example: 135
- $1^1 = 1$
- $3^2 = 9$
- $5^3 = 125$
- Sum = $1 + 9 + 125 = 135$

Test Cases:
1.175->Disarium number
2.123->Not a Disarium number
3.0->Invalid input

Your Code

```cpp
#include <iostream>
#include <cmath>
#include <string>
using namespace std;
class DisariumChecker {
private:
    int num;
public:
    DisariumChecker(int n) {
        num = n;
    }
    void checkDisarium() {
        if (num <= 0) {
            cout << "Invalid input" << endl;
            return;
        }
        string numStr = to_string(num);
        int length = numStr.length();
        int sum = 0;
        for (int i = 0; i < length; i++) {
            int digit = numStr[i] - '0';
            sum += pow(digit, i + 1);
        }
        if (sum == num) {
            cout << "Disarium number" << endl;
        } else {
            cout << "Not a Disarium number" << endl;
        }
    }
};
int main() {
    int n;
    if (cin >> n) {
        DisariumChecker checker(n);
        checker.checkDisarium();
    }
    return 0;
}
```

Input

```
135
```

Output

```
Disarium number
```

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QUESTIONS

Q1
Check whether a number is a sunny number
5 marks

Q2
Check whether a number is a fascinating number
5 marks

Q3
Check whether a number is a Disarium number
5 marks

Q4
Check whether a number is a Kaprekar number
5 marks

Q5
Check whether a number is a Keith number
5 marks

Q6
Check whether a number is a Trimorphic number
5 marks

6/6 answered

Q4 Check whether a number is a Kaprekar number

5 marks

Topics to use: Classes, objects, member functions, loops, digit counting, arithmetic operations, conditional logic.

Question: A positive number is a Kaprekar number if its square can be split into two parts such that the sum of those parts equals the original number. The right part must contain as many digits as the original number.

Example: 45
- Square = \(45^2 = 2025\)
- Split into 20 and 25.
- Sum = \(20 + 25 = 45\).

Test Cases:
1.45->Kaprekar number
2.55->Kaprekar number
3.10->Not a Kaprekar number

Your Code

```cpp
#include <iostream>
#include <string>
using namespace std;
class KaprekarChecker {
    int num;
public:
    KaprekarChecker(int n) : num(n) {}
    void checkKaprekar() {
        if (num <= 0) {
            cout << "Invalid input" << endl;
            return;
        }
        if (num == 1) {
            cout << "Kaprekar number" << endl;
            return;
        }
        string s = to_string((long long)num * num);
        bool ok = false;
        for (int i = 1; i < s.length(); i++) {
            long long l = stoll(s.substr(0, i));
            long long r = stoll(s.substr(i));
            if (r > 0 && l + r == num) {
                ok = true;
                break;
            }
        }
        cout << (ok ? "Kaprekar number" : "Not a Kaprekar number") << e
    }
};
int main() {
    int n;
    if (cin >> n) KaprekarChecker(n).checkKaprekar();
    return 0;
}
```

Input

```
45
```

Output

```
Kaprekar number
```

Visible Test Cases

Test Case 1 ✓

Test Case 2 ✓

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- Q5 Check whether a number is a Keith number — 5 marks
- Q6 Check whether a number is a Trimorphic number — 5 marks

6/6 answered

Q5 Check whether a number is a Keith number — 5 marks

Topics to use: Classes, objects, member functions, loops, arrays, digit extraction, sequence generation.

Question: A Keith number is a positive integer whose digits form the beginning of a sequence. Each subsequent term is the sum of the previous terms, using as many terms as the original number has digits. If the sequence reaches the original number, it is a Keith number.

Example: 197

Starting digits: 1, 9, 7
- \(1+9+7=17\)
- \(9+7+17=33\)
- \(7+17+33=57\)
- \(17+33+57=107\)
- \(33+57+107=197\)

Therefore, 197 is a Keith number.

Test Cases:
1.742->Keith number
2.14 ->Keith number
3.123->Not a Keith number

Your Code

```cpp
#include <iostream>
#include <vector>
#include <string>
using namespace std;
class KeithChecker {
    int num;
public:
    KeithChecker(int n) : num(n) {}
    void checkKeith() {
        string s = to_string(num);
        int len = s.length();
        vector<int> seq;
        for (char c : s) seq.push_back(c - '0');
        while (true) {
            int nextTerm = 0;
            for (int i = seq.size() - len; i < seq.size(); i++) {
                nextTerm += seq[i];
            }
            if (nextTerm == num) {
                cout << "Keith number" << endl;
                return;
            }
            if (nextTerm > num) {
                cout << "Not a Keith number" << endl;
                return;
            }
            seq.push_back(nextTerm);
        }
    }
};
int main() {
    int n;
    if (cin >> n) KeithChecker(n).checkKeith();
    return 0;
}
```

Input

```
742
```

Output

```
Keith number
```

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- Q6 Check whether a number is a Trimorphic number — 5 marks

6/6 answered

Q6 Check whether a number is a Trimorphic number — 5 marks

Topics to use: Classes, objects, private data members, member functions, loops, arithmetic operations, remainder operator.

Question: A number is called a Trimorphic number if its cube ends with the original number.

Example: 24
- Cube = \(24^3 = 13824\)
- The last two digits are 24.
- Therefore, 24 is a Trimorphic number.

Test Cases:
1.24->Trimorphic number
2.7->Not a Trimorphic number
5.-5->Invalid input

Your Code

```
#include <iostream>
#include <string>
using namespace std;
class TrimorphicChecker {
    int num;
public:
    TrimorphicChecker(int n) : num(n) {}
    void checkTrimorphic() {
        if (num < 0) {
            cout << "Invalid input" << endl;
            return;
        }
        long long cube = (long long)num * num * num;
        string sNum = to_string(num);
        string sCube = to_string(cube);
        if (sCube.length() >= sNum.length() && sCube.substr(sCube.lengt
            cout << "Trimorphic number" << endl;
        } else {
            cout << "Not a Trimorphic number" << endl;
        }
    }
};
int main() {
    int n;
    if (cin >> n) TrimorphicChecker(n).checkTrimorphic();
    return 0;
}
```

Input

```
24
```

Output

```
Trimorphic number
```

Visible Test Cases

- Test Case 1 ✓
- Test Case 2 ✓
- Test Case 3 ✓

5.00 / 5 marks

Submitted